Counterfactual Cell World: A Controlled Testbed for Single-Cell Perturbation Prediction

Atila Vahedian

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Abstract

This note describes a small research system for predicting how a cell population changes under unseen perturbations. The first version uses a synthetic single-cell world with known regulatory structure, hidden combinatorial interventions, and distribution-level evaluation. The point is not to claim biological discovery. The point is to build a hard, inspectable counterfactual benchmark before moving to real Perturb-seq data.

1 Problem

For each condition, the model observes a source population $X \in \mathbb{R}^{n \times g}$, a perturbation vector $p \in \mathbb{R}^{g}$, a cell type c, a dose d, and a time value t. It predicts a target population $Y \in \mathbb{R}^{n \times g}$.

The difficult case is a held-out perturbation combination. The model may see single perturbations or other combinations during training, but it must predict the target distribution for a combination that was never directly observed.

2 Model

The model has four pieces:

1. a signed learned gene interaction matrix A,
2. a context encoder over population statistics, perturbation, cell type, dose, and time,
3. a residual latent transition applied to source cells,
4. a Gaussian decoder that predicts per-cell expression mean and variance.

The objective is

$$\mathcal{L} = \mathrm{NLL}(Y \mid \mu, \sigma) + \lambda_{\mathrm{mmd}} \mathrm{MMD}(\mu, Y) + \lambda_1 \|A\|_1 + \lambda_{\mathrm{dag}} h(A).$$

The likelihood term rewards cell-level accuracy. The MMD term rewards population-level agreement. The graph penalties keep the learned map sparse and discourage cyclic shortcuts.

3 Synthetic World

The simulator samples latent gene programs, cell-type offsets, sparse regulatory couplings, and perturbable genes. A target cell population is generated by integrating a nonlinear regulatory drift:

$$x_{k+1} = x_k + \Delta t \left[-0.32 x_k + \alpha \tanh(x_k A^\top + b_c + 0.25p + 0.95pA^\top) \right] + \epsilon.$$

The stronger indirect term makes direct expression shifting a weak baseline for held-out combinations.

4 Current Result

method	MSE	MAE	MMD	gene corr.
model	0.0386	0.1422	0.1021	0.8714
direct shift	0.1090	0.2356	0.1711	0.7218
mean shift	0.1428	0.2692	0.1905	0.7271

Table 1: Held-out combinatorial perturbation split from the checked synthetic run. Lower is better except gene correlation.

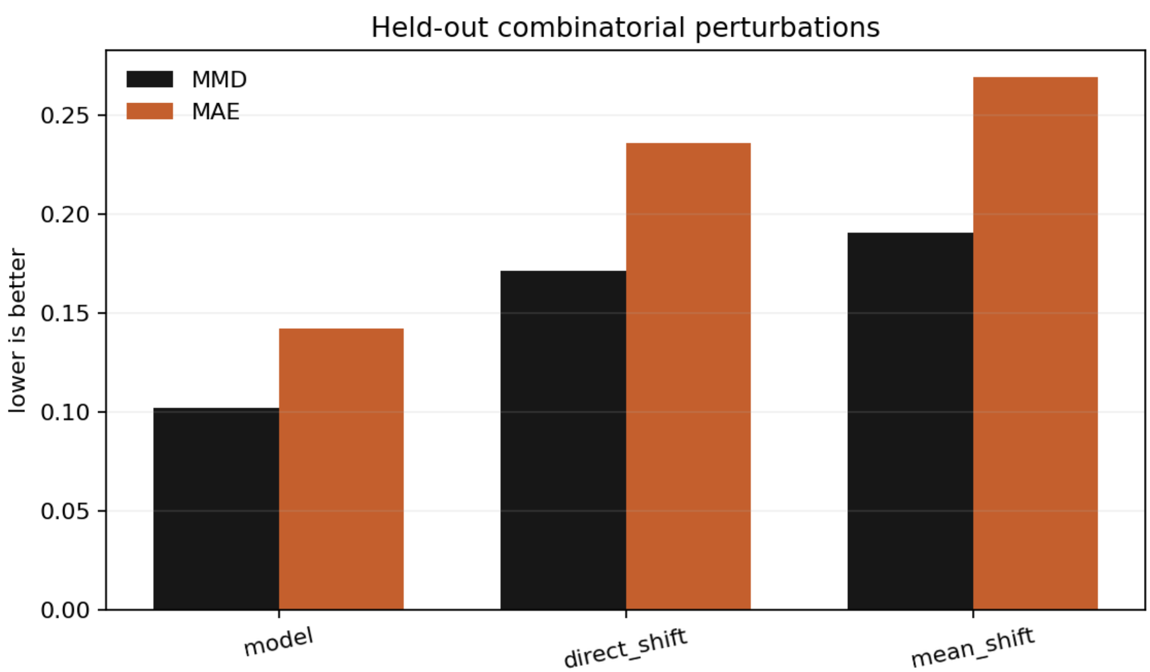

Figure 1: Held-out metric comparison against two simple baselines.

5 Next Step

The next serious version should freeze a real Perturb-seq benchmark and report the same metrics by cell type, intervention order, dose, and unseen-gene setting. Public single-cell atlases are available through CELLxGENE, but atlas data and perturbation data should not be mixed casually because tissue, donor, assay, and batch effects can dominate the learning problem.